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An Exploratory Study of Minor League Baseball Statistics

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An Exploratory Study of Minor League Baseball Statistics

Gabriel Chandler and Guy Stevens

Abstract

We consider the problem of projecting future success of Minor League baseball players at each level of the farm system. Using tree based methods, in particular random forests, we consider which statistics are most correlated with Major League success, how Major League teams use these statistics differently in handling prospects, and how prior belief in a players ability, measured through draft position, is used throughout a players Minor League career. We show that roughly the 18th round prospect corresponds to being draft neutral for a team, whereas teams essentially make decisions based strictly on performance. We use for our data all position players drafted between 1999 and 2002.

KEYWORDS: random forests, baseball, classification

1 Introduction

Whether a prospect ultimately has success in Major League Baseball (MLB) is a difficult achievement to predict; sometimes, individuals who appear on the fast track to success never make the highest level, while relatively unknown players surprise everyone and become superstars. Brien Taylor and Steve Chilcott were both the top overall draft pick in their respective drafts, yet neither ever played in a single Major League game. On the other hand, Mike Piazza, considered one of the greatest catchers of all time, was drafted in the 62nd round (for perspective, the 2012 draft consisted of only 40 rounds).

Almost all prospects, obtained as high school or collegiate athletes through the Rule 4 Amateur Draft each June, spend time in the Minor Leagues (MiLB), honing their skills and adjusting to the lifestyle and quality of competition in professional baseball. Each of the 30 MLB organizations has a system of Minor League affiliates through which their players progress. While players often skip individual levels according to their own development, the typical progression remains the same; Rookie Class, Class A Short-Season, Class A, Class A Advanced, Double-A, and Triple-A. Depending on age and prior experience, players may begin their professional career at any of these levels. An organization must monitor each player's progress throughout his career to decide when he is ready to be promoted to the next level, or when he is no longer on track for big-league success and should be released.

Even with standard performance statistics being available on Minor League performance, players who do reach the Major Leagues frequently fail to produce at an acceptable level. Both Spurr and Barber (1994) and Longley and Wong (2011) considered factors related to progression through the ranks for Minor League pitchers. Spurr and Barber (1994), using data from mid 1970s to the late 1980s, found that performance relative to the mean dictated how quickly a pitcher

moved to the next level. Longley and Wong (2011) found that Minor League statistics do not hold strong predictive power over Major League performance, and a heavy reliance on performance early in a pitcher's career can lead to the early release of a player with Major League potential. Pitchers are notoriously volatile, prone to injury or sudden lack of success. We wonder here if hitters, less injury prone than pitchers, may be more predictable and therefore safer investments.

Prospects are an increasingly important commodity in baseball. More and more teams have begun to focus on player development instead of big-money free agent signings. Drafted and developed talent is cheap relative to the price of free agents; once reaching the Majors, these players are initially paid the league minimum and remain under team control with relatively minor salary increases for up to six seasons, often providing value far beyond what they are paid. As a result, a good deal of literature exists on analyzing draft strategies. For instance, Shughart and Goff (1992) showed that playing at a four-year college is related to a quicker progression through the Minor Leagues. Molitor and Winfree (2007) found that lower round high school draftees can maximize their expected earnings by choosing to go to college. Spurr (2000) concluded that there is no statistically significant difference between clubs in their ability to find Major League prospects via the draft. This suggests that there is a large deal of uncertainty in projecting the development of players. Due to this uncertainty, young stars often fail spectacularly rather than reach the lofty expectations of their clubs.

As small-market organizations become increasingly dependent on successes from their Minor League ranks, the cost of mismanaging these prospects is growing rapidly. Teams have incredible incentives to handle their young talent as effectively as possible; because these players can provide substantial surplus values at the big league level, any advantage in projecting Major League success can be a huge boon

for the team. As Burger and Walters (2009) point out, while there is a low probability of draft picks ultimately achieving Major League success, the large expected returns suggests that any slight advantage a team might garner could be substantial. Cheaply obtaining other team's prospects that are likely to outperform their club's valuation is another possible way to strengthen the club's development track. For example, former Reds' GM Jim Bowden defended his policy of scouting prospects that would be drafted by other teams by explaining, "when their stock goes down, we're going to trade for them" (Sullivan, 2001).

The goal of the current investigation is two-fold. We examine the problem of predicting eventual Major League success by looking at a player's career trajectory through the Minor Leagues. Players often fail to make the Majors entirely, which can be a costly failure for the club due to escalating signing bonuses. As found by Longley and Wong (2011), for pitchers, this is not an easy prediction to make. In the present work, we instead focus our attention on position players and their offensive abilities. Statistics that would be most useful are those that have strong predictive power at all levels, which would allow reliable judgements to be made at the lower Minor League levels. As it seems obvious that statistics at more advanced classes should be more predictive of Major League success, we seek to quantify this.

Secondly, we examine how different organizations use information about prospects, which might suggest inefficiencies in the evaluation process. This includes not only performance statistics at the various levels, but also their more subjective measurement of talent which we quantify by draft position.

We mention at the outset that this is an exploratory investigation into offensive MiLB statistics, an area that has not been explored in a level of detail even approaching that of MLB, due in part to the difficulty in accumulating such data. Analyses of prospects and predictive power of their performances are common in other sports as,

for instance, NCAA basketball and football have much greater visibility than MiLB (see e.g. Coates and Oguntimein, 2008, who looked at NCAA basketball performance and its relation to NBA success). While there is undoubtedly much to discover through detailed analysis, we present an overview of the sort of information that can be gleaned from such data. The following section contains a description of the data that was used. Section 3 contains an overview of the techniques we used in analyzing the data, most notably random forests. The results of our analysis are found in section 4, with a discussion in the section following.

2 Data

Our data comes from The Baseball Cube (www.thebaseballcube.com), a website with an extensive database of statistics from both MLB and MiLB. Draft team data was taken from the web site My MLB Draft (www.mymlbdraft.com).

The data set consists of all hitters who were drafted and signed by a Major League team between 1999 and 2002, and played in at least 15 minor league games. These players are drafted between the ages of 17-23, depending on their experience. Many are drafted after graduating high school, while others have college experience. The years that we selected reflect a desire to use the most current data available which also allows for a final evaluation of whether a player has reached the Major Leagues or not. Our particular time frame was also chosen to reflect evaluation of talent that is consistent with the paradigm shift associated with the release of “Moneyball” (Lewis, 2003). An additional limiting factor is the difficulty in attaining this type of data. We none-the-less have over 1000 ($n=1019$) players who played at least 15 games at Class A.

The data for each player is separated into an array of statistics at each of the six Minor League levels. Table 1 contains the variables,

Stat	Description	Min	Mean	Max	SD
AB	At Bats	8	830.851	3753	662.548
AVG	Batting Average	0.091	0.264	0.391	0.04
OBP	On-Base Percentage	0.161	0.334	0.455	0.042
SLG	Slugging Percentage	0.091	0.408	0.737	0.091
OPS	On-Base Plus Slugging	0.303	0.743	1.131	0.127
ISO	Isolated Power	0	0.144	0.427	0.064
ISO/BA	Isolated Power by AVG	0	0.538	1.377	0.217
R/AB	Runs per At Bat	0	0.141	0.256	0.033
DBL/AB	Doubles per At Bat	0	0.054	0.25	0.02
TPL/AB	Triples per At Bat	0	0.006	0.029	0.006
HR/AB	Home Runs per At Bat	0	0.025	0.093	0.018
RBI/AB	Runs Batted In per At Bat	0	0.13	0.269	0.044
SB/AB	Stolen Bases per At Bat	0	0.021	0.144	0.024
SB%	Stolen Base Success Rate	0	0.567	1	0.28
BB/AB	Walks per At Bat	0.016	0.099	0.244	0.035
SO/AB	Strikeouts per At Bat	0.062	0.211	0.545	0.063
SO/BB	Strikeouts per Walk	0.533	2.456	12	1.434
HBP/AB	Hit-by-Pitch per At Bat	0	0.012	0.057	0.009
GIDP/AB	Double Plays per At Bat	0	0.023	0.103	0.012

Table 1: Summary of Triple-A Statistics.

including a summary of each statistic at the Triple-A level. These statistics measure different aspects of each player's offensive performance. We assume the reader is familiar with most of these statistics, save perhaps isolated power (ISO), which is calculated by subtracting batting average from slugging percentage. This stat is aimed purely at determining the hitters proficiency at getting extra-base hits (doubles, triples, and home runs), predominately a power metric. Defensive ability, while undoubtably associated with a player's evaluation, is not included in our study as data of reliable defensive metrics does not exist for Minor League baseball. Future improvements on this sort of analysis can be made by taking into account both position and defensive ability.

The response we created, ML Contribution, was a binary vari-

able. A value of 1 meant the player established himself in the Major Leagues (which we define as having played over 320 games through the 2011 season). This somewhat arbitrary definition of success corresponds to roughly two full seasons of Major League experience. It does not account for varying levels of success, and future research should explore different metrics of success. Our choice of data ensured that players who were going to develop into successful MLB players would have had time to do so.

Because many players follow a somewhat unique route through the Minors, as some skip a level along the way and many fail to reach higher levels, we divided the data set into six subsets. Each subset corresponds to one of the six Minor League levels, containing only the players who played at least 15 games at that level.

We note that all of the statistics under consideration are fairly standard, though the list is fairly comprehensive in the realm of classical baseball statistics. In recent years, many new statistics have been proposed, falling under the umbrella of ‘sabermetrics.’ However, much of the information required for calculating these statistics is not available for Minor League games. When such statistics were computable, for instance OPS or ISO, we included it. Thus, the choice of variables that we considered in this study is comprised of essentially all classical offensive statistics as well as those modern statistics that are computable from the available data. We are also working under the assumption that no player is released who otherwise would have become a successful Major League player. This seems a reasonable assumption, as many minor league players are kept on well past the point when an organization has lost hope on a player in order to fill roster spots.

We also note that the start of these players professional careers coincides in an era when performance-enhancing drug (PED) use was believed to be fairly common. There is no way to know which players, and how many, were using PEDs, though it is possible that this

can contaminate a significant amount of recent data. While there is no doubt that PED use causes issues on an individual level, we suggest that when looking across the entirety of MiLB, teams are still using the same criteria in evaluating players regardless of whether or not their achievements have been “enhanced.” This issue may add extra uncertainty into our model in the case of players who start to use PEDs midway through their career, making their earlier statistics unreliable in predicting Major League success. We present our analysis with an acknowledgement of this concern.

To check the generalization of our results to other years, the New York Mets organization allowed us access to a data set of a selection of draftees by all clubs from 1995-1998, the four years immediately preceding our window. Though not comprehensive, it allowed us to consider the question of whether evaluation methods have changed over time, albeit in a fairly short time frame. The variables in this data set are identical to those used in the primary one.

3 Classification Trees and Random Forests

The foundation of our statistical analysis is the Classification Tree (see, for instance, Hastie, Tibshirani and Friedman, 2001), which uses any number of explanatory variables to predict a categorical response variable by a series of splits. Beginning with the entire set of data at the starting node, this method constructs a tree by dividing this node into two splits based on a condition on a single statistic (for example, $AVG > .250$ and $AVG \leq .250$). Then, at each of the subsequent nodes, it is split again the same way based on any of the statistics in the set. This continues until the tree is well developed and sufficient in accomplishing the goal of “purity.” For instance, the oft-used Gini

impurity is defined at the i^{th} node as

$$I_G(i) = \sum_{j=1}^m q_{ij}(1 - q_{ij})$$

where q_{ij} is the proportion of observations in the i^{th} node of the j^{th} class. Here, $m=2$ (either Major League success or not or promoted beyond the current level or not). In creating the tree, enough splits are made so that most nodes are very pure. However, the pruning process that follows removes any over-specification, allowing for some impurity but maintaining the usefulness of the results. In the end, each player belongs to a single “terminal node” and a single probability of success for all observations within the node.

Building on the usage of the Classification Tree is the Random Forest technique (Breiman, 2001). This is an ensemble classifier that creates a large number of trees, each randomly selecting a subset of variables to use at each split in the tree and limiting consideration for the split to this subset. Additionally, each tree is grown based on a bagged (bootstrapped) sample of the training data. By comparing the results of the numerous tree iterations, the Random Forest algorithm can determine which variables were most significant. This is accomplished in two ways. One way is by taking the “out of bag” (OOB) observations (those not used to grow the tree) and comparing the number of correct classifications against the number of correct classifications when we permute the values of this variable. Alternatively, one can look at the reduction in impurity when the variable is used, which is what we use in the present work. The average of these quantities is called the *variable importance*, V . It quantifies how useful a variable is for prediction of the response. For an individual player, their estimated probability of success, $\hat{p}$, is computed as the proportion of trees in the Random Forest which predict a 1 for the response when that players statistics are “pushed down” each tree in the forest. Furthermore, a measure of tree accuracy is given by

the OOB error rate, which is the misclassification rate of observations that were not used to grow the tree, thus avoiding a biased error rate due to over fitting.

The Random Forest technique has been shown to be fruitful in analyzing sports data, for instance in predicting baseball Hall of Fame voting (Mills and Salaga, 2011). Random Forests are well-suited for these situations as they can combine a wide range of variables in unexpected, non-linear relationships.

While some of our variables are highly correlated (for instance SLG and OPS), Archer and Kimes (2008) have shown through an extensive simulation study that variable importance based on Gini impurity works well in a variety of settings in identifying influential predictors, including when covariates are correlated. Additionally, we considered the method of Diaz-Uriarte and Alvarez de Andrés (2006) for variable selection, which uses backward stepwise elimination to achieve a set of “important” covariates. This was done using the associated R package *varSelRF* (Diaz-Uriarte, 2010).

4 Results

We analyzed trends across Minor League levels using various slope analyses. Two outputs of the Random Forest lent themselves very nicely to this type of analysis: the variable importance score V_{ij} for variable i at Minor League level $j = 1, \dots, 6$, and the estimated probability of promotion for individual players, $\hat{p}_k$, $k = 1, \dots, n$. We used the evolution of V_i over levels to compare both the predictive power of performance metrics and the relative usefulness of draft position in promoting prospects. The Random Forests were run in R (R Development Core Team, 2011), using the *randomForest* package of Liaw and Wiener (2002). For each Random Forest, we used standard values of the various parameters. Each run consisted of 500 trees (which was sufficient for the OOB error rate to roughly converge), and we used

the standard rule of thumb of $\sqrt{k}$ variables at each iteration, where k is the number of co-variates. In our setting $k=19$ and thus we used 4 variables at each iteration. This choice of k was competitive with others in terms of OOB error rates. To quantify the performance of the Random Forests, the OOB error rates are provided in Table 2. Not surprisingly, there is a positive trend in the error rate as we move up through the levels, as the probability of a random selected player reaching the Majors naturally has a greater probability (closer to .5) at higher levels than at lower ones. At the lowest levels, one can be quite accurate in terms of overall error rate by predicting no one to become a productive Major Leaguer. In fact, for players who played at least 15 games in Rookie Class, only 47 of the 857 had Major League success according to our definition. At AAA, where such a naïve strategy would fail, the Random Forest is fairly accurate. This table gives the first indication that predicting Major League success is both easier done at higher levels as well as very difficult to do.

Level	Rookie	Low A	A	High A	AA	AAA
Sample Size	857	852	1019	931	664	443
OOB Error Rate	0.0548	0.048	0.074	0.091	0.137	0.167
False Negative Rate	0.000	0.001	0.006	0.009	0.044	0.073
False Positive Rate	1.000	1.000	0.925	0.867	0.635	0.480

Table 2: Sample sizes and error rates for the Random Forests at each level of MiLB.

To explore the generalization of our results to other time periods (with hopes that they hold for modern day), we pushed the individuals contained in our smaller 1995-1998 data set down our trees and calculated their ensemble prediction (whichever outcome is predicted by the majority of the trees in the Random Forest). The class that consisted of the most observations was AAA, with 251 observations, 104 of whom went on to have Major League success (41 percent vs 23 percent in our primary data set, suggesting that this is not a random sample of AAA players). For this data set, we found a false

positive rate of 62.5 percent and a false negative rate of 6.8 percent.

4.1 Projecting Major League success based on performance statistics

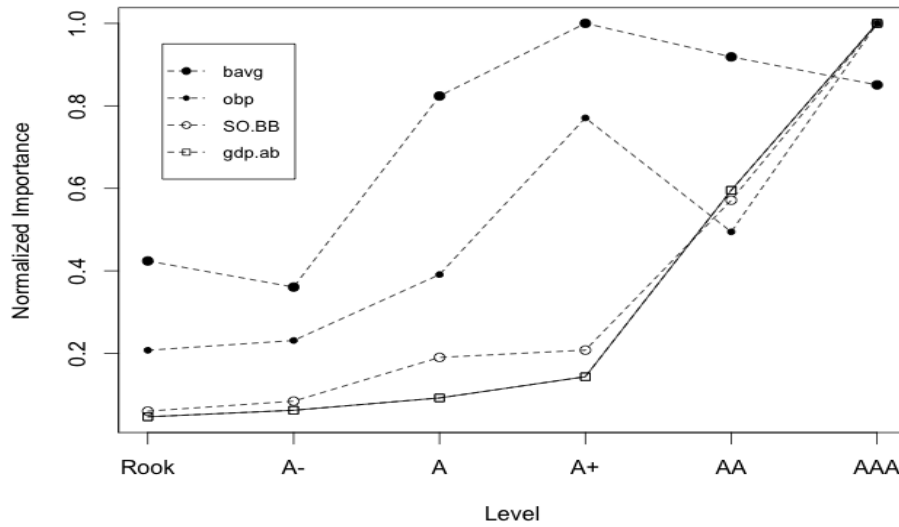


Figure 1: Normalized importance values at each level for four statistics.

We took the subset of players who made it at least as far as Triple-A and created a Random Forest, including the statistics from all six Minor League levels. The importance values of each statistic are greater for the higher levels, as expected.

In order to compare importance trends across statistics, we computed normalized importance values

$$\tilde{V}_{ij} = \frac{V_{ij}}{\max_j V_{ij}},$$

i.e. normalized according to the maximum value for each statistic. For most statistics, this was the AAA importance, yet there were a few

exceptions, which is likely due to sampling variability. Using these normalized values, we quantified the evolution of the importance by calculating a slope for each of the $\tilde{V}_i$. In order to calculate a slope, we had to determine the “distance” between each level. Both a graphical analysis as well as an application of the Box-Cox method suggested that assigning consecutive integers to successive levels was suitable. Figure 1 shows an example of the trends found in some statistics.

Figure 2 shows these slope values plotted against their importance values at AAA only. The graph area might meaningfully be divided into four quadrants, with variables falling into each quadrant having similar properties. The bottom-left corner, corresponding to low-importance, low-slope statistics, contains statistics with very little predictive power at any level, though tending to be stable for each player. As expected, GDP/AB and HBP/AB fall into this category. More surprising is that BB/AB also fit this description, as walk rate is typically considered a very useful metric in other areas.

The bottom-right corner, corresponding to low-importance, high-slope statistics, contains statistics whose low-level prediction value increases in value with progress through the Minor Leagues. However, even their explanatory power at Triple-A is not particularly impressive. This includes RBIs, which sabermetricians have almost universally agreed upon is a statistic not truly determined by player ability but rather by team strength and luck, which suggests its value as a future predictor is likely quite small.

The top-right corner corresponds to high-importance, high-slope statistics. These statistics are those whose value at the higher Minor League levels, like Double-A and Triple-A, is notable, yet still carry very little predictive power in the lower levels. These include the typical rate statistics that analysts like to use: AVG, OBP, SLG, OPS. While the results still suggest that these are not reliable for low level performance, these are the most predictive statistics as prospects get closer to the Major Leagues.

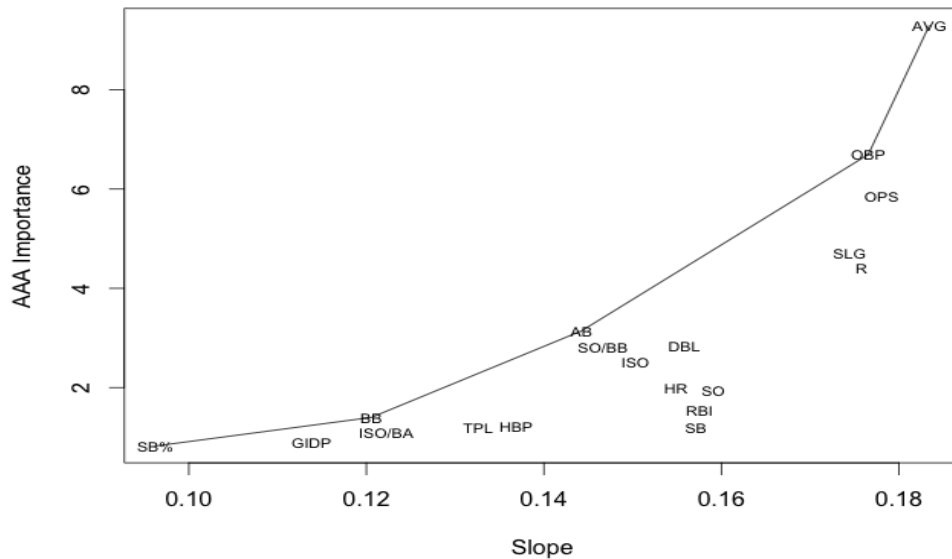


Figure 2: Importance of AAA statistics versus Slope of importance.

Finally, the top-left corner corresponds to high-importance, low-slope statistics. These would be statistics that have strong predictive power of Major League success at every level of the Minor Leagues. Such statistics would be very valuable to Major League clubs, as they would allow decision-makers to make more accurate conclusions about prospects earlier in their Minor League careers. The lack of any such statistics suggests that definite conclusions about a player based on his performance in the low Minors are not easy; Minor League performance does not begin to distinguish prospects' future Major League performance until the higher levels.

We conjecture that there is a convex relationship that relates variables importance to their slope. This could suggest some constraint on decision-making that prevents smart decisions early. To visualize this, we have plotted the least convex majorant to our particular variables. Sabermetric statistics may push this boundary outward,

but likely not all the way to the target area. In general, the major impact of sabermetric analysis is a better judgment of a player's value to his team, not predicting how his statistics against inferior competition will correlate with those at the highest level. The problem we face is that early conclusions are faulty, requiring promotion to higher levels to really put together a prospect's statistical portfolio. Somewhat unsurprisingly, the variables selected via the backward stepwise method of Diaz-Uriarte and Alvarez de Andrés (2006) were all standard metrics, ISO/BA, OPS, and AVG.

This does not mean that Rookie ball statistics are completely useless. By creating a subset of the data including players in the first 315 picks of each draft, corresponding to about 10 rounds, we find that certain statistics are significantly more important than others for certain players. Notably, for these early picks, Strikeouts per At Bat and Strikeouts per Walk are far more significant than others. Many of these players are potential-rich high school picks, those with the physical tools for success but who lack experience against higher-level pitching. If these players are overwhelmed in Rookie ball, leading to high strikeout rates, it is a very poor indicator for their Major League career.

For this subset, the importance values level out for all stats except the standard rate metrics by High A. At this level and at Double-A, OPS is far more important than any other statistic in determining Major League success. At Triple-A, AVG, OBP, and OPS carry similar importance values, while all other statistics are much less significant. The same trends are not present for those drafted lower than pick 315, which show no outliers in terms of importance. Essentially, top draft picks that display the type of performance that their potential indicated was possible can distinguish themselves in the high minors, as shown in the traditional rate statistics. Unfortunately, lesser draft picks do not appear to have this opportunity. Our statistics seem to fail to illuminate success-bound players taken in the lower rounds of

the draft.

4.2 How draft pick number is used

The second piece of our analysis relates to the power of draft status in a prospect's progression through the Minor Leagues. The response variable is whether or not a player was meaningfully promoted beyond his current level (defined as playing at least 15 games at a higher level than that considered, as often prospects will be promoted for shorter time periods due to extenuating circumstances like injuries). Using draft pick number in addition to performance variables, we can analyze the motivators to promote a given prospect, specifically how influential draft position is to an organization's evaluation of a player. We suspect that a highly touted pre-draft prospect is likely to be promoted from lower levels regardless of his statistics. This corresponds to our previous results showing that low level statistics do not project well for later success. Likewise, we conjecture that late-draft players will require better performances in lower levels to justify (or perhaps establish) their promise.

Perhaps the most fundamental question is how good scouts are at recognizing talent pre-draft and how unpredictable players development is. To this effect, we calculated the correlation between a players draft position (in terms of pick number) and their OPS at each level. A negative correlation indicates that players drafted earlier tended to have higher OPS. While this is verified by the data, the numbers are fairly low (see table 3).

Rookie	Low A	A	High A	AA	AAA
-0.116	-0.169	-0.142	-0.249	-0.123	-0.124

Table 3: Sample correlation values between draft number and OPS at each level.

To analyze how 'prior' information about a player factors into

promotion decisions, we looked at how the likelihood of promotion was affected by the inclusion of draft pick in our analysis. We created a Random Forest for each level-specific subset of data, obtaining the estimated probability of meaningful promotion based on only the player's statistics at that level. Here, the estimated probability is the proportion of trees in the Random Forest that predict promotion for the terminal node that the player belongs to. We also created a second Random Forest with the same statistics and subset, additionally including the player's draft pick number, DP . This yielded a second estimate of the probability of promotion, which we denote as $\hat{p}_{DP}$.

We then calculated the difference between these two probabilities—representing the change in likelihood of promotion based on the inclusion of draft pick—and regressed this value on the player's position in the draft, i.e. we fit the model

$$\hat{p} - \hat{p}_{DP} = \beta_{0j} + \beta_{1j}DP + \epsilon.$$

We analyzed this regression across each level-specific subset, looking at the resulting slopes $\hat{\beta}_{1j}$. Unsurprisingly, for many players, the change in estimated probability is essentially 0, corresponding to the fact that draft position correlates with performance itself. However, as we can see, players who were drafted higher (smaller values for draft position) are often given the benefit of the doubt at this low level. Conversely, players who are not highly regarded and thus drafted lower are often much less likely to be promoted given their statistics. The change in probabilities based on the inclusion of draft number can be quite dramatic, with values exceeding 20 percent in either direction. We note that the x-intercept for the line of best fit occurs at draft position 549 (somewhere around Round 18). This seems to represent the point at which an organization is neutral to draft position and seemingly relies solely on empirical evidence of ability.

As expected, there is a decreasing trend in the slopes as we move up through the Minor Leagues, with the slope at the Rookie

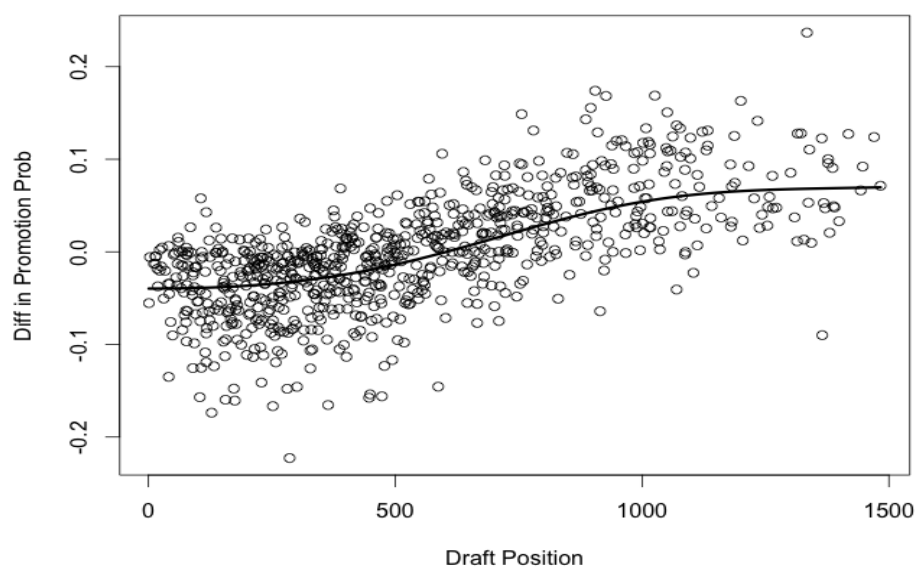


Figure 3: Difference in promotion probability versus draft position at Low A with loess curve (solid line).

level being 2.75 times larger than at AAA. This signifies that actual performance becomes more important to an organization's evaluation of talent as level increases.

As we noted at the end of 4.1 above, Rookie level statistics do have some value to clubs. While they are not indicative of Major League success, there is information to be gleaned from these statistics nonetheless. Still using the difference in probabilities, we took the residuals from a fitted line for each observation in the Rookie data set. This residual gives a measure of performance, in terms of probability of promotion, relative to some expected performance level. Positive residual values indicate a player outperforming expectations, while negative values correspond to underperformance.

Logistic regression of these residuals on Major League success returned a positive coefficient with a one-sided p-value of .027, showing statistically significant evidence that Rookie performance relative to expectations is related to probability of Major League success. Again, our earlier analysis showed us that Rookie statistics are not very strong predictors of future success, however they can be used to gain information about players. In this case, Rookie ball performance significantly different than expectations should lead a club to reevaluate their opinion of a player. While vastly outperforming expectations does not make the player a lock for future success, clubs may want to consider viewing a player differently than they did prior to Rookie ball. A 30th round pick who plays like a 1st rounder in Rookie ball should not suddenly be viewed as a top tier prospect, but his club's valuation of him may increase beyond what his draft status would suggest. The same can be said in the opposite direction; while a first round pick who plays like a 30th rounder should not become an afterthought, his club may view him as a less-promising prospect than it previously had. Organizations do not need to make finite conclusions based on these early performances, but they might consider shifting their initial beliefs and projections nonetheless.

4.3 Differences in organizational philosophies

The final section of our analysis looked into what factors each franchise uses in evaluating its Minor League players. We ran a separate Random Forest on each organization's Minor League system, evaluating the chance of promotion at each Minor League level. As earlier, this gave us an importance value for each statistic and each team at each level. We normalized these by each statistic's maximum in order to compare them across teams and levels.

At each Minor League level, we took the mean importance value and the variance of the importance values for the 30 Major League organizations. A sample of this for three select statistics is shown in Figure 4. These three stats show very interesting trends worth further analysis.

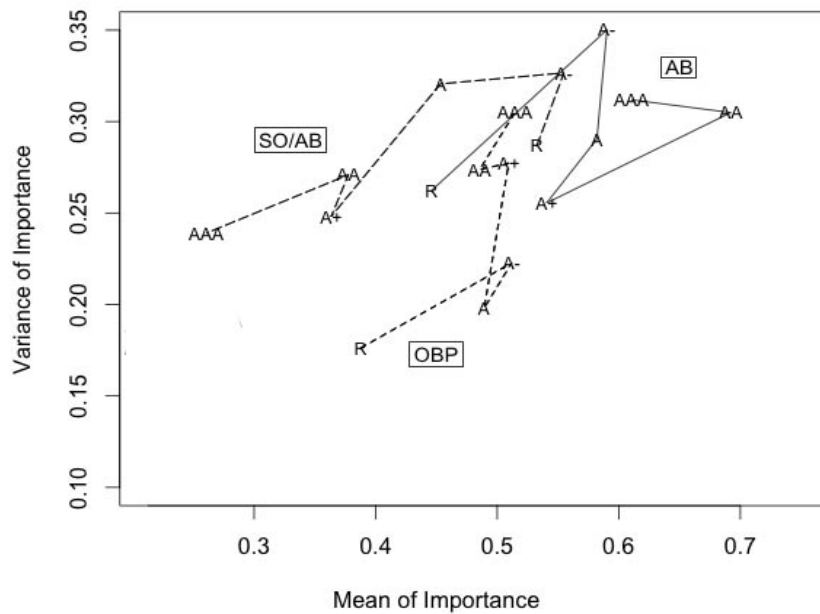


Figure 4: Importance to promotion across Major League organizations.

OBP has a very clear trend across the six Minor League levels. While not monotonically increasing in mean and variance, its gen-

eral path is that. This means that, for higher levels, OBP becomes a more and more important factor in the promotion of a player. At the same time, teams disagree more and more, showing a divide in teams' methodologies. After Michael Lewis' bestseller *Moneyball* (2003), OBP became a widely used statistic in judging a player's offensive worth, yet it did not catch on with all teams. As some teams were resistant to this change, the data shows significant differences in how teams value OBP. The Athletics, the focus of *Moneyball*, fittingly showed the highest importance values for OBP at the higher levels. The Red Sox and Blue Jays, also known for their early implementation of statistical analysis and belief in *Moneyball* ideas, showed similarly high importance values for OBP and OPS.

SO/AB shows the reverse trend, decreasing in importance across levels. We found earlier that strikeout rate is meaningful at the lower levels for early draft picks, and this high mean and variance for Rookie and Low A seem to suggest that this had been implemented by some teams. The decreasing trend in importance means, once a player has been promoted through the ranks, as long as his other production is there, an accompanying high strikeout rate will not prevent him from being promoted. The simple fact that he has been promoted to a higher level suggests that he is performing in other areas (a high slugging percentage for example).

AB may be the most telling of this group. At the higher levels, number of at bats carries the highest mean importance values of any statistic, while also having substantial variance. This suggests that, while teams disagree on how to value at bats, it is widely used as the single most important statistic in promoting players. This says that teams may have predetermined the players they consider their most promising prospects, and thus give them the most at bats to prepare them for the next level. Furthermore, we suggest that, rather than using their AA and AAA teams to evaluate numerous players and promote those who are successful, they have a handpicked group

of prospects they are evaluating ahead of the rest regardless of, and possibly despite, their performance numbers.

Leading the pack of AB-heavy franchises is the Pittsburgh Pirates, an organization who saw very little in terms of recent success in player development despite consistently desirable draft position due to a lack of Major League success. Their organization has been loaded and reloaded with incoming impact talent, yet rarely had a Pirates prospect become an impact player at the highest level until the 2012 season. The Chicago Cubs' development track, also having a noteworthy dependence on at bats, has been unable to provide Major League-caliber players on a regular basis. Finally, the New York Yankees provide an interesting case. Due to their large budget, they have less incentive to develop players who will actually be successful at the Major League level. Much of the value they derive from player development is the ability to trade prospects who are considered valuable by other clubs. This might explain a reliance on at bats, promoting the players they believe will be valuable trade chips to other clubs.

5 Discussion and Future Directions

The results of our study suggest, unsurprisingly, that Minor League hitters cannot be accurately judged by their performances early in their Minor League careers. Some conclusions seem to be possible early on – for example, strikeout rates in the low Minors are important for high draft picks – but a player's overall body of work does not become truly telling of Major League potential until he has progressed to the higher levels. The apparent functional upper bound on statistics' predictive power might be viewed as a challenge to analysts; the identification of a statistic pushing this boundary would provide a legitimate advantage to evaluators. It is of interest to see where on this graph (Figure 2) modern statistics favored by sabermetricians would fall. As mentioned earlier, such statistics are generally

not available from MiLB.

We also analyzed the effect of draft pick number in the development and promotion of a prospect. We saw that higher picks are more likely to be promoted at any given set of performance statistics, proving that performance is not the only tool used by clubs to decide the worth of a prospect. Instances of promotions of underperforming early round draft picks are common across MiLB. However, as a prospect progresses toward the Major Leagues, his performance rightly begins to outweigh his draft location in the eyes of his club. Whether this happens quickly enough is an interesting question. For instance, Jackson Williams, a catcher taken in the first round in the 2007 draft by the San Francisco Giants has consistently played over undrafted Tyler LaTorre throughout their MiLB careers, despite LaTorre having a much higher career OBP (.343 vs .307), as well as their 2011 season together at AAA (.331 vs .288).

The most telling aspect of the draft pick analysis is the evidence that performance relative to expectations is important. An unknown prospect who dominates early on could be seen as a legitimate prospect, while early failures of a top prospect are meaningful negative signals for his future career. While nothing can be said definitively based on low level MiLB performance, there appears to be a great deal of uncertainty in the process that leads to suboptimal player development.

Our team-by-team analysis suggested some interesting insights into clubs' decision-making. On-base percentage, strikeout rate, and home run rate all showed interesting trends in franchises, while teams' clear preferences for their chosen prospects seem to be overshadowing actual performance in the high Minors, shown in the valuation of at bats. This might be seen as a meaningful inefficiency, as emphasis on at bats seemingly does not match its unimpressive predictive power over future Major League success.

An important question is how decision-making changes over

time. The introduction of the *Moneyball* era, for instance, should be readily apparent through a similar analysis on much earlier data. Future work should consider whether results from a decade or so ago are still valid. As the success of today's Minor Leaguers is still unknown, this question is a difficult one to answer.

Another question of interest to MLB clubs that we did not address is how a player should be promoted through the various levels towards MLB. While we assumed that players were not released until it was certain that they would not develop into a successful Major Leaguer for their club, it is possible that players careers were affected by the particular schedule they followed. It is often worried that promoting a player too quickly, especially younger players, can hamper their development. This may be reflected in the importance that AB has for some clubs. While the appropriate promotion schedule is likely individual specific, it would be interesting to see if any general trends exist.

In the speculative world of player development, any marginal advantage in decision making can be parlayed into long-term success. Our exploratory analysis suggests that such advantages can be found. Future analyses should incorporate defensive ability, as well as position, as this is certainly being taken into account by the individual organizations. One might expect varying organizational emphases on defense, which, given the disagreement over the importance of fielding ability (see *Moneyball*), may lead to advantages for clubs who value it appropriately.

The ultimate success of a Minor League prospect depends on many factors, only a few of which have been accounted for here. While we've already noted sabermetrics, position, and defensive ability, injury and need at the MLB level are two additional variables that were left out of our model. These are difficult, as the occurrence of career ending injuries is often not reported, and would lead to censored data. MLB need is difficult to quantify. Improvements in our study

can be made by bringing in any of these factors that were unavailable to us.

References

- Archer K.J., and Kimes, R.V. (2008). Empirical characterization of random forest variable importance measures. *Computational Statistics & Data Analysis*, **52(4)** 2249–2260.
- Breiman L. (2001). Random Forests. *Machine Learning*. **45**, 5-32.
- Burger, J.D., and Walters, S.J.K. (2009). Uncertain prospects: rates of return in the baseball draft. *Journal of Sports Economics*, **10** 485–501.
- Coates, D. and Oguntimein, B. (2008). The length and success of NBA careers: does college production predict professional outcomes? *North American Association of Sports Economists Working Paper Series*, **08-06**.
- Cohen, G, (2011). The Baseball Cube. www.thebaseballcube.com, 15 June 2011.
- Diaz-Uriate, R., and Alvarez de Andrés, S. (2006). Gene Selection and Classification of Microarray Data Using Random Forest. *BMC Bioinformatics* **7:3**.
- Diaz-Uriarte, R. (2010). varSelRF: Variable selection using random forests. R package version 0.7-3. <http://CRAN.R-project.org/package=varSelRF>
- Hastie, T., Tibshirani, R., and Friedman, J.H. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction, New York: Springer.

- Lewis M.L., (2003). *Moneyball: The Art of Winning an Unfair Game*. Norton: New York.
- Liaw, A. and Wiener, M. (2002). Classification and Regression by randomForest. *R News* **2**, 18–22.
- Longley, N. and Wong, G. (2011). The speed of human capital formation in the baseball industry: the information value of minor-league performance in predicting major-league performance. *Managerial and Decision Economics*, 32: 193–204.
- Mills, B.M. and Salaga, S. (2011). Using Tree Ensembles to Analyze National Baseball Hall of Fame Voting Patterns: An Application to Discrimination in BBWAA Voting, *Journal of Quantitative Analysis in Sports* Vol. 7: Iss. 4, Article 12.
- myMLBdraft, 15 June 2011 www.myMLBdraft.com.
- R Development Core Team (2011). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, <http://www.R-project.org/>.
- Shugart, W. F., II, and Goff, B. L. (1992). Field of dreams: On the construction of professional baseball talent in colleges and the minor leagues . In G. W. Scully (Ed.), *Advances in the economics of sport* (Vol. 1, pp. 91-114). Greenwich, C: JAI Press.
- Spurr, S.J. and Barber, W. (1994). The effect of performance on a worker's career: evidence from minor league baseball. *Industrial & Labor Relations Review*, **47**, 692–708.
- Spurr, S.J. (2000). The baseball draft: A study of the ability to find talent. *Journal of Sports Economics*, **1**, 66–85.

- Sullivan, T. (2001). Bowden makes deals - and enemies, *The Cincinnati Enquirer*. http://reds.enquirer.com/2001/04/01/red_bowden_makes_deals_-.html 2 Dec. 2011.
- Winfrey, J.A. and Molitor, C.J. (2007). The value of college: Drafted high school baseball players. *Journal of Sports Economics*, **8**, 378–393.